VOLUME IV

Platform Engineering

Purpose

Volume IV defines the production engineering architecture of Project APEX. It specifies backend services, infrastructure, storage, APIs, security, deployment, observability, and operational practices required to transform the research platform into an enterprise-grade system.

Engineering Vision

The platform is designed as a modular, cloud-native, event-driven system with clear service boundaries, strong security, scalability, and extensibility. Each subsystem can evolve independently while sharing common data models and platform services.

Volume IV Structure

Chapter 23 — Backend Microservices

Service decomposition and bounded contexts.

Chapter 24 — API Gateway

Routing, authentication, rate limiting, and aggregation.

Chapter 25 — Event Bus & Messaging

Asynchronous communication and event-driven workflows.

Chapter 26 — Storage Architecture

Relational, vector, graph, object, and cache storage.

Chapter 27 — Authentication & Authorization

Identity, RBAC, OAuth, API keys, and permissions.

Chapter 28 — Plugin SDK

Extension framework for desktop and application plugins.

Chapter 29 — Deployment Architecture

Containers, orchestration, CI/CD, and environments.

Chapter 30 — Monitoring & Observability

Logging, metrics, tracing, alerting, and diagnostics.

Chapter 31 — Security Architecture

Threat model, encryption, secrets, auditing, and compliance.

Chapter 32 — Scalability & Reliability

High availability, resilience, disaster recovery, and performance.

Expected Outcomes

- Production-ready backend architecture
- Secure and scalable platform services
- Extensible plugin ecosystem
- Cloud-native deployment model

• Enterprise observability
• Operational excellence